Structural certificates for $\mathrm{PSp}_4(3)$ and its extension

Private L054 finite application

Theorem 1. *Relative to the explicitly acquired numerical classifications and named constructor interface, all 116 rows of the current* `psu42` *table and all 350 rows of* `pgammau42` *have the displayed structural descriptions. These are existence assertions for the actual row groups, not assertions that a structural label determines a unique group.*

Exact source and embedded inputs. The current PDFs, attributed to Thomas Connor and Dimitri Leemans, display the ambient names $\mathrm{PSp}_4(3)$ and $\mathrm{PSp}_4(3) : 2$. Their 21 native pages, including the final caption-only page of the first table, were directly read. The source filenames are locators, not extra group-identification assumptions. All retained subgroup arrays are literal `Group` expressions in permutations, on 40 and 85 points respectively. We acquire their complete accepted numerical inventories and exact source-row bijections; no proper group is reconstructed from a printed name.

The article's left-association convention applies except where native parentheses prescribe a factor. In particular, row 4 of `psu42` and row 12 of `pgammau42` print $3^{1+2} : (Q_8 : 3)$, whereas row 20 of the latter prints $3^{1+2} : (Q_8 : 2)$. These parentheses are retained. A source 3^3 is elementary abelian of order 27, not a cyclic group of order 33. Dots, colons and crosses retain the acquired nonsplit, nontrivial split and direct meanings. The unsigned extraspecial and Q_{12} conventions are inherited with their proofs. We make no historical-byte identity claim about these current PDFs.

One explicit named construction. GAP Reference Manual 50.2-15 defines `PSp(4,3)` as the projective symplectic group preserving a nondegenerate alternating form, with default permutation action on projective lines. We retain this construction identity as an explicit software interface. The original successful numerical run has exactly this constructor in its driver, a clean terminal receipt and unchanged recorded inputs and outputs. Its actual top generators equal the 40-point generators imported here. This concrete group is the named atom; its name is not deduced from its order 25,920.

The second original numerical model was produced as a permutation image of the automorphism group of $\mathrm{PSU}_4(2)$. Its displayed symplectic extension label is paid by a new actual decomposition: the checker verifies a normal subgroup of order 25,920, an explicit isomorphism from the named PSp atom, and an actual order-two complement with nontrivial action. Separately, its source row 2 is identified with the same atom by an explicit map. Thus the source label does not follow merely by comparing orders or by assuming an exceptional unitary–symplectic isomorphism.

Registry reuse. There are 129 acquired concrete structural models and this one named input. The new certificate contains 38 models: 31 split, five direct and two nonsplit. Repeated names may select different actual models. Every factor and occurrence uses an explicit map to its embedded target. The accepted MC.3 generator-edge criterion is used unchanged: a bijection fixing identity and respecting every right multiplication by each ordered generator is a homomorphism by induction on generator words, hence an isomorphism.

Proof of all row assertions. The literal verifier closes every finite permutation group. References to new factor models are acyclic. In a split extension it verifies the actual normal subgroup N, the certified structural map to N, the actual complement K with its own certified map, $N \cap K = 1$, and $|N||K| = |H|$. Hence $H = NK$. Normality is checked on generators and extends to their generated groups. A noncommuting generator pair verifies the nontrivial action; in a direct model, all pairs of factor generators commute, which proves that the factors commute elementwise.

One new nonsplit model has normal subgroup $3 : S_3$ and quotient S_3. The acquired quotient-lift criterion first checks fixed quotient generators xN, yN, the relations $x^2, y^3, (xy)^2 \in N$, and all six distinct cosets. A complement exists exactly when some $a \in Nx$, $b \in Ny$ satisfy $a^2 = b^3 = (ab)^2 = 1$. Every one of the 324 lift pairs is tested and fails. The other new nonsplit model has normal subgroup $3^2 : S_3$ and quotient C_2; all 54 outside elements fail to have order two. These exhaustive literal checks prove nonsplitting. An unsuccessful software complement search is never used as proof.

Every final source occurrence is matched to the exact acquired subgroup under the numerical source-row map. The source literal, native syntax tree, order, actual record number and all generator arrays are checked. The certificate contains 540 explicit isomorphisms and 382,991 checked right-generator edges. Together with the new decompositions and acquired atom identities, these checks prove all 466 assertions in the theorem.

Candidate and verification evidence. One bounded GAP candidate pass completed every row in 15.69 seconds. The normal-subgroup, complement and isomorphism operations only propose witnesses. The final replay uses literal arithmetic and does not repeat that discovery, any ambient subgroup census or an automorphism calculation. Three physical line continuations in candidate JSON occur at the already accepted digit–digit or comma–digit positions. The unchanged numeric reader removes only these continuations, parses strict JSON, and validates all resulting permutation data. The sole initial development failure was an invalid loop in a preparer adaptation, rejected by Python's parser before writing any input or executing GAP; its exact bytes are retained.

The full-certificate damages test incorrect identity or generator edges, missing elements and source rows, altered subgroup assignments and generators, powers and dots, the named constructor model, wrong complements, circular references and an actual S_3 quotient lift. Separate interface controls require literal imported arrays, preserve the parenthesized factor, reject malformed syntax and nonnumeric continuation positions, and exercise the actual retained-constructor acquisition with incomplete, wrong-model, failed and changed-input receipts. Ordinary and optimized Python must yield identical complete reports.

Scope. This application adds 466 current structural row joins. The numerical classifications, original execution identities, earlier model proofs and named PSp constructor are explicit acquired inputs. No canonical uniqueness of extension notation, historical file identity, new classification algorithm, whole-original admission, question resolution, formalization or measurement credit is claimed. The application and exact source joins are authored here under root's registry-reuse task; distinct review remains separate.